



BANNAGA ALTIEB ABDUL MUHSIN

Electrical Project Engineer

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Location

Riyadh, Saudi Arabia

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EDUCATION

B.Sc. Electronic Engineering (Control)

El Neelain University

Khartoum, Sudan · 2007–2012

CERTIFICATIONS

- CEM® — Certified Energy Manager
- CMVP® — Measurement & Verification
- LEED Green Associate
- Machine Learning with Python
- Power BI Essential Training
- SCADA / PLC / DCS / RTU

LANGUAGES

Arabic — Native

English — Professional

MEMBERSHIPS

Saudi Council of Engineers (SCE)

Sudanese Engineers Association (SEA)

AVAILABILITY

Sun–Thu: 8:00 AM – 6:00 PM

Weekends: on-call for urgencies

Open to short-notice travel

PROFILE

Certified Energy Manager (CEM®) and Certified Measurement & Verification Professional (CMVP®) with 14+ years of experience in electrical and energy engineering. Specializes in project management, telecom power systems, energy-efficient solutions, and technical operations across diverse infrastructure. Expertise spans data collection, energy analysis, facilities management, AI-driven dashboards, and structured reporting — delivering innovative, reliable, and performance-driven engineering solutions across telecom, power generation, and industrial sectors. Fluent in English and Arabic.

WORK EXPERIENCE

Electrical Project Engineer

2019 – Present

ACES | Riyadh, Saudi Arabia

- Serve as Energy Manager and efficiency designer across STC COW and Mobily ENM managed networks.
- O&M Engineer for Mobily ENM Managed Services — full facility operations, CAFM/BMS integration, and KPI reporting.
- O&M Engineer for STC COW Managed Services — operations and maintenance across 600+ nationwide telecom sites.
- Delivered multi-phase energy optimization programs achieving up to 18% reduction in utility costs.
- Developed Python automation for energy reporting and ML-based anomaly detection alerts.
- Built an AI-driven Energy & Environmental Dashboard for CO2 footprint monitoring across all STC COW sites.
- Mentored junior engineers and standardized field engineering practices.

Electrical Project Engineer

2018 – 2019

Absar | Riyadh, Saudi Arabia

- Executed SEC unified-contract builds for 60+ MV/LV electrical assets (RMUs, transformers, LV switchgear).
- Performed full commissioning, testing, and zero-defect SEC handover documentation.
- Installed and configured SCADA monitoring and control system integration.

Electrical Engineer

2017 – 2018

BEMCO

- Preventive and corrective maintenance for 11kV distribution networks.
- Developed standardized testing, inspection, and commissioning manuals.
- Managed NCR resolutions and maintained technical documentation workflows.

Electrical Engineer

2012 – 2017

SEC Thermal Generation | Sudan

- Day-to-day operations and maintenance for a 200 MW thermal generation block.
- Managed wastewater treatment systems using PH meters and OC 4000 DCS controls.
- Preventive maintenance and fault management for 11kV distribution systems.

KEY PROJECTS

STC COW Network — O&M & Energy Optimization

End-to-end O&M for 600+ telecom sites nationwide; systematic energy auditing, SCADA oversight, and multi-phase optimization achieving up to 18% utility cost reduction.

Mobily ENM Managed Services

Full O&M and facility operations including CAFM/BMS integration, outage prevention, and structured performance reporting.

Mobily Data Center Energy Audit

Comprehensive energy audit of critical data center infrastructure delivering measurable PUE improvements, certified under CEM® / CMVP® methodology.

SEC Unified Contract — MV/LV Commissioning

Installation and commissioning of 60+ MV/LV assets with full SCADA integration and zero-defect SEC handover.

PP12 Thermal Generation Plant — 200 MW

Operations including 11kV switching, safety routines, wastewater treatment, and DCS-based environmental monitoring.

AI CO2 & Energy Dashboard

ML-based anomaly detection and CO2 footprint tracking dashboards built with Python, Power BI, SQL Server, and DAX.

SKILLS & EXPERTISE

Energy Management (CEM® / CMVP®)



SCADA / DCS / PLC / RTU



Python & Automation / ML



Power Systems — HV / MV / LV



Power BI & Data Analytics



Project & O&M Management



KEY ACCOMPLISHMENTS

- Reduced nationwide telecom site utility costs by up to **18%** through multi-phase energy optimization.
- Delivered uninterrupted O&M coverage for STC COW & Mobily ENM with proactive outage prevention.
- Cut incident resolution time by **30%** via troubleshooting guides and automation routines.
- Commissioned **60+ MV/LV assets** under SEC unified contracts with full SCADA integration.
- Built an AI-driven Energy & Environmental Dashboard tracking real-time CO2 emissions and efficiency.
- Modeled operational trends across 600+ multi-site networks using Power BI, Python, and SQL Server.

AI & DEVELOPMENT

AI & Automation

- Designed Python automation to streamline energy performance reporting and KPI tracking.
- Developed ML-based anomaly detection alerts for telecom site power consumption.

- Built governance-ready Energy & Environmental dashboards for CO2 footprint monitoring.
- Applied Power BI, SQL Server, and DAX to model operational trends across multi-site networks.

Development & Tools

- Created reusable troubleshooting guides and diagnostic scripts for SCADA / PLC / DCS systems.
- Documented API-style checklists to standardize field data capture and site integrations.
- Version-controlled documentation and engineering templates using Git/GitHub workflows.

"You never change things by fighting the existing reality. To change something, build a new model that makes the old one obsolete." — R. Buckminster Fuller

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